AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	LightEdge Austin II, TX -- station KAUS, America/Chicago
Committed pair	OSM 379204643 -> 383888101 LightEdge Austin II / Lumen Austin 1
Facade gap	596.2 m between the two halls
Weather record	43,746 real hourly records from KAUS
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.1848 C at 40 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-01-02 (safe_sector)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 6 of 24 hours with 1 mode change. The reactive incumbent took 15 hours -- 9 more -- but declared 0 unsafe hours against the agent's 0. 12 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	6 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	15 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 12 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	-	5.001	18.0	6.110	5.045
01	mechanical	yes	-	5.561	18.0	5.560	4.434
02	mechanical	yes	-	6.111	18.0	5.560	3.950
03	mechanical	no	refusal	6.113	18.0	5.560	3.330
04	mechanical	yes	switch budget	6.660	18.0	5.000	3.357
05	mechanical	yes	switch budget	6.671	18.0	2.220	2.860
06	mechanical	yes	switch budget	6.722	18.0	0.560	2.420

07	mechanical	no	refusal	7.781	18.0	1.110	2.706
08	mechanical	yes	-	11.111	18.0	6.110	3.579
09	mechanical	yes	-	14.451	18.0	8.890	4.720
10	mechanical	no	refusal	17.221	18.0	11.110	6.061
11	mechanical	yes	-	16.671	18.0	13.890	6.640
12	mechanical	no	refusal	16.671	18.0	15.560	7.033
13	mechanical	yes	-	16.671	18.0	16.670	6.431
14	mechanical	yes	-	17.780	18.0	17.780	4.001
15	mechanical	no	refusal	18.331	18.0	17.780	3.481
16	mechanical	yes	-	17.781	18.0	17.780	2.732
17	mechanical	no	dry-bulb	18.342	18.0	11.110	2.971
18	FREE	yes	-	17.781	18.0	8.890	3.341
19	FREE	yes	-	16.670	18.0	8.330	3.665
20	FREE	yes	-	16.112	18.0	3.890	4.024
21	FREE	yes	-	15.561	18.0	2.220	4.585
22	FREE	yes	-	15.560	18.0	4.440	5.070
23	FREE	yes	-	9.440	18.0	5.560	5.322

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00   MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.001 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00   MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.561 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00   MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.111 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00   MECHANICAL   [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

04:00   MECHANICAL   [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.660 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.671 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 6.722 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

08:00 MECHANICAL
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.111 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.451 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

11:00 MECHANICAL
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.671 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

12:00 MECHANICAL [refusal]
Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

13:00 MECHANICAL
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.671 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more.

This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

14:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.781 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.781 C against a 18.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.342 C against a 18.0 C limit, so it fails by 0.342 C. It would take a limit 0.342 C higher, or a bound 0.342 C tighter, to change this hour.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.781 C, which is 0.219 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.341 C, and the margin is measured, not chosen: 3.341 C of group-conditional forecast error for this hour of day, plus 0.0007 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.670 C, which is 1.330 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.665 C, and the margin is measured, not chosen: 3.665 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.112 C, which is 1.888 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.024 C, and the margin is measured, not chosen: 4.022 C of group-conditional forecast error for this hour of day, plus 0.0022 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.561 C, which is 2.439 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 4.585 C, and the margin is measured, not chosen: 4.584 C of group-conditional forecast error for this hour of day, plus 0.0006 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 15.560 C, which is 2.440 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.070 C, and the margin is measured, not chosen: 5.069 C of group-conditional forecast error for this hour of day, plus 0.0004 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.440 C, which is 8.560 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 5.322 C, and the margin is measured, not chosen: 5.321 C of group-conditional forecast error for this hour of day, plus 0.0002 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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